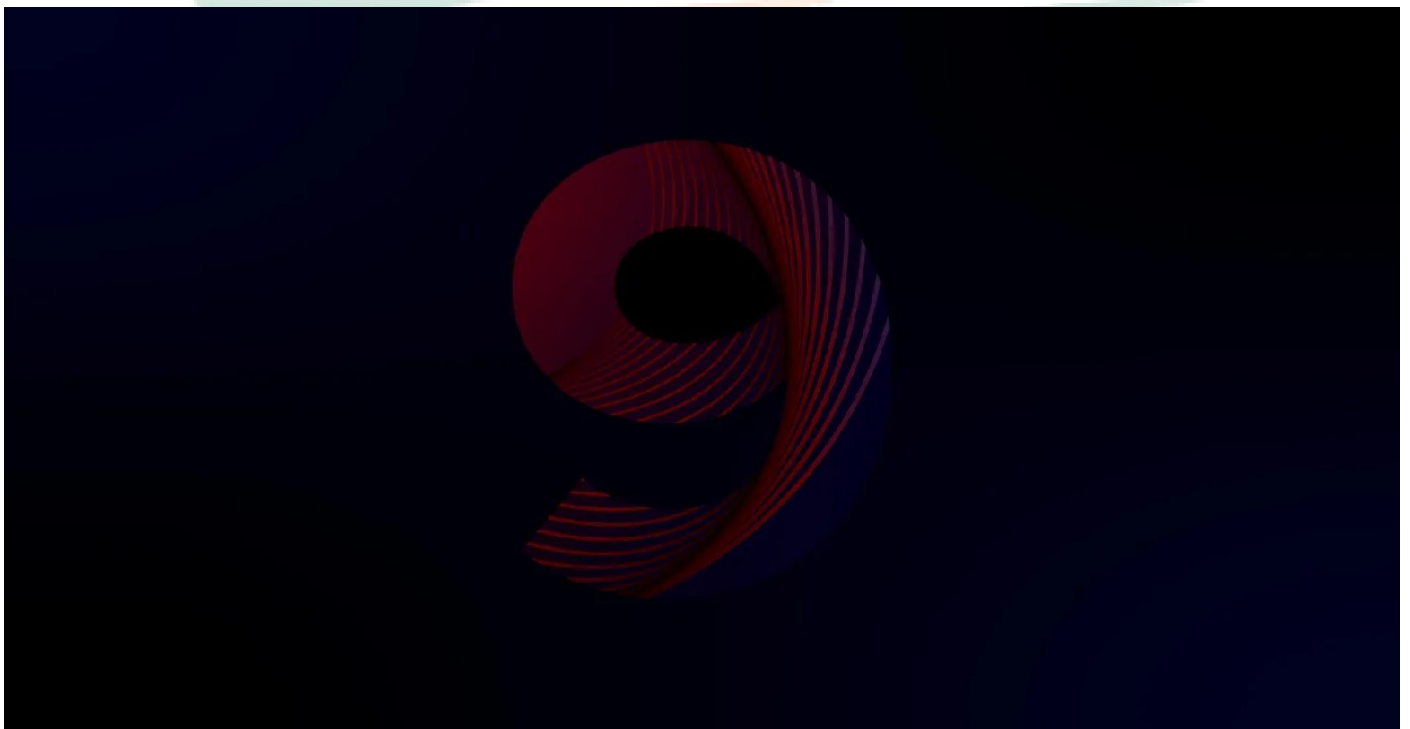


RHCSA EXAM QUESTIONS

EX200 RHEL 9



<https://www.ipsr.org>

1. Exam hall / Exam view

2. RHCSA Sample Questions

RHCSA EXAM SCORE

Manage basic networking: 100%

Understand and use essential tools: 100%

Operate running systems: 100%

Configure local storage: 100%

Create and configure file systems: 100%

Deploy, configure and maintain systems: 100%

Manage users and groups: 100%

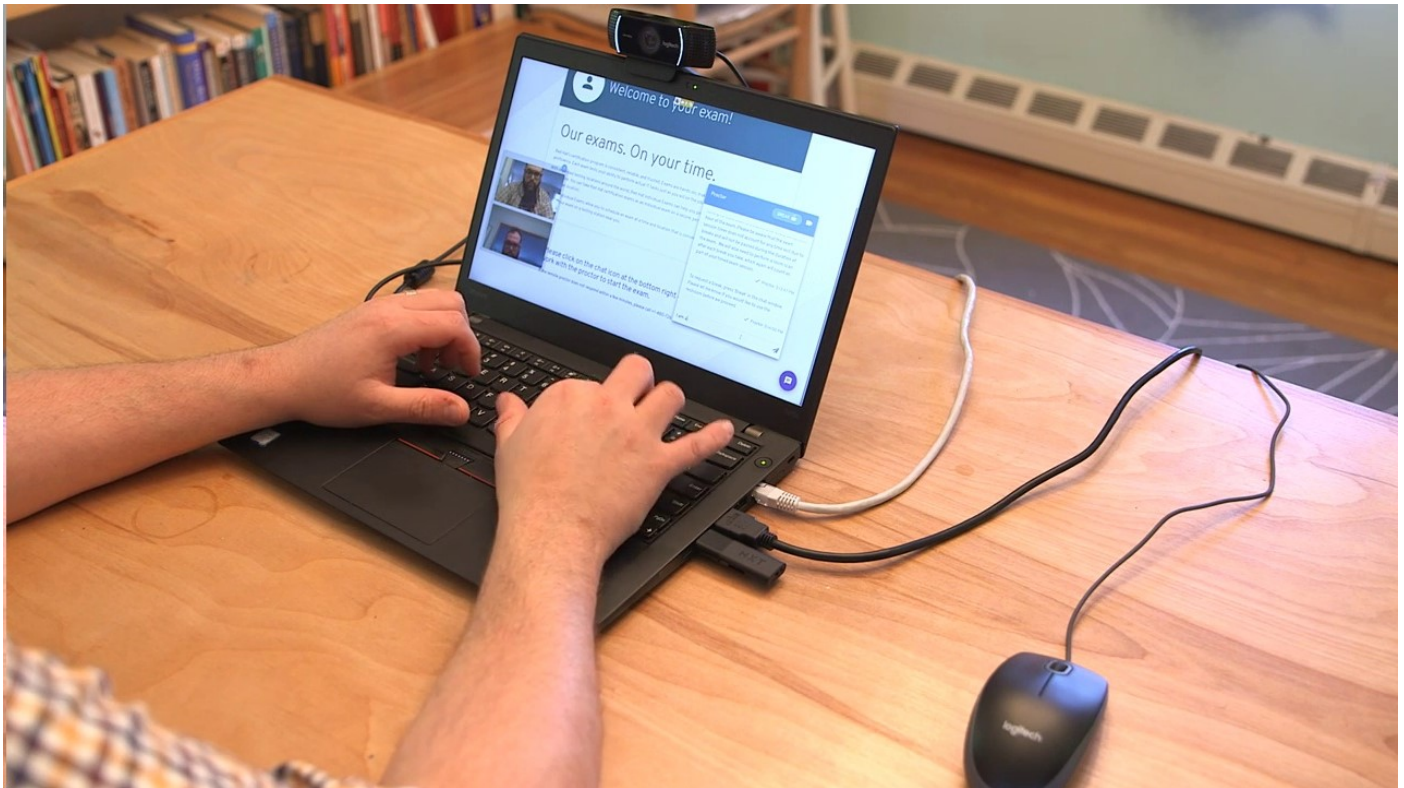
Manage security: 100%

Manage containers: 100%

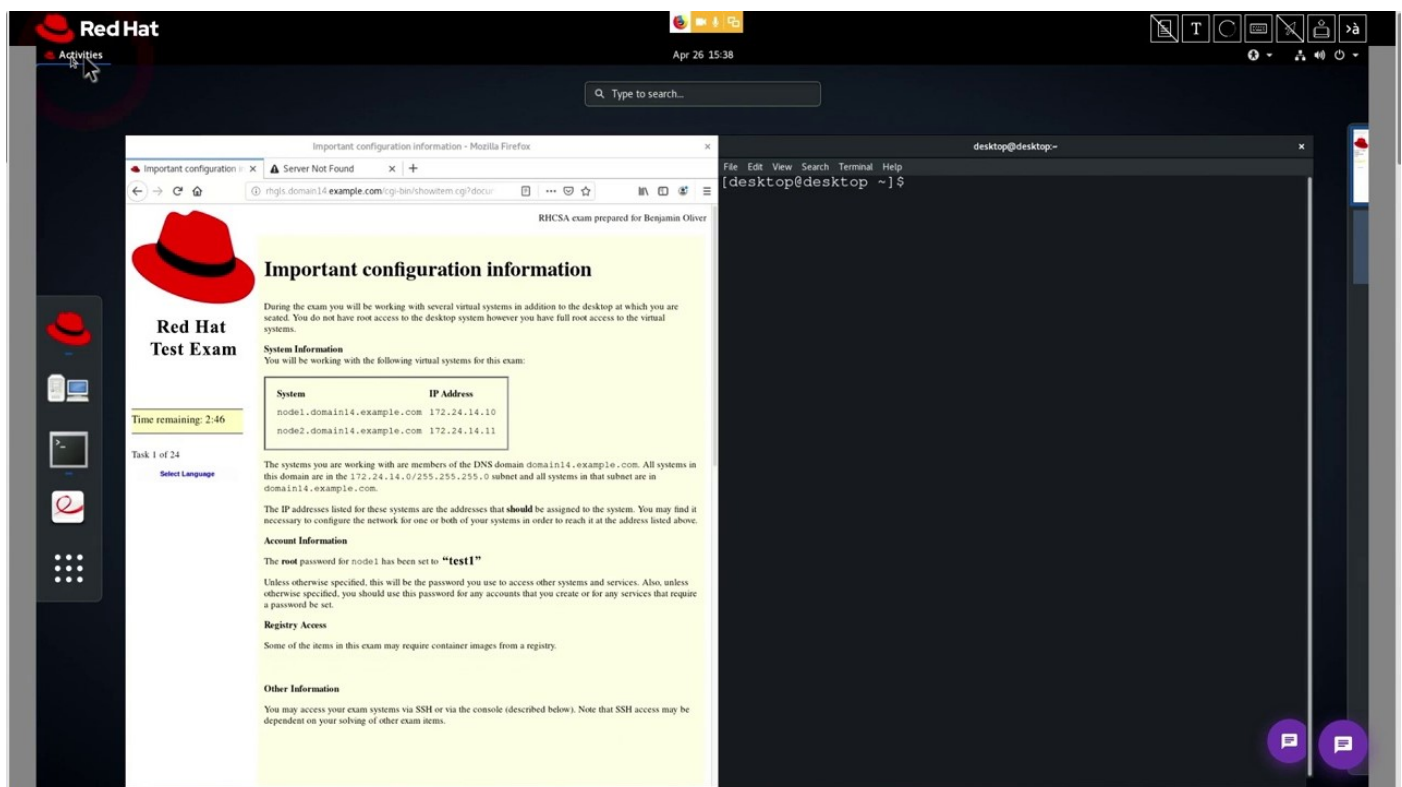
Create simple shell scripts: 100%

Exam hall / Exam view

1. Get started RHCSA Exam



2. Exam view / click activities / Read carefully





Exam View

VM Monitor

Terminal

Exam View

**Red Hat**

Activities View Exam

Apr 26 16:36

Red Hat Certified System Administrator Exam - Mozilla Firefox

Red Hat Certified System Administrator Exam

Index of /BaseOS

rhgs.domain14.example.com/cgi-bin/exam.cgi#

**Red Hat
Test Exam**

Time remaining: 1:49

Select Language

Perform the following tasks on node1.domain14.example.com.

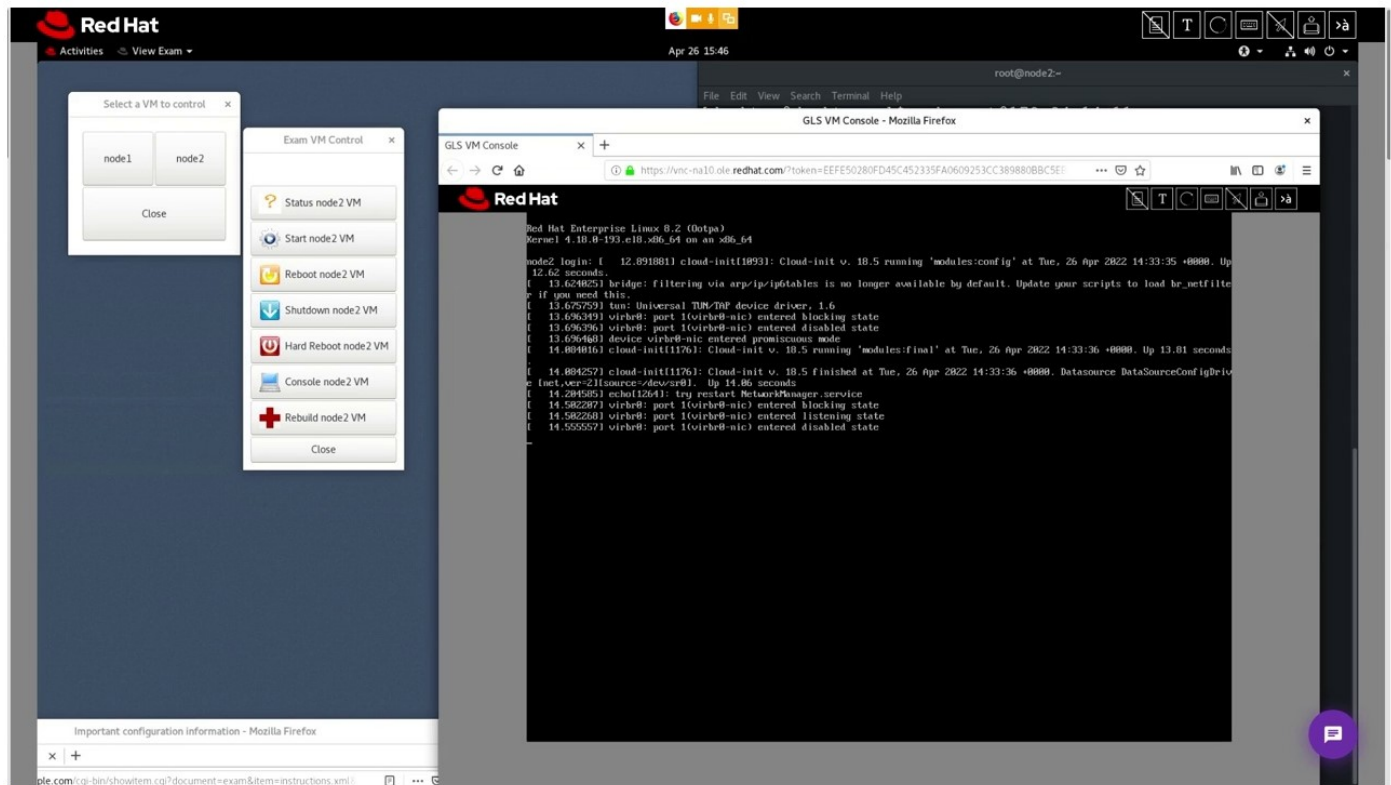
☐ Revisit	☐ Done	Configure your system 01
☐ Revisit	☐ Done	Configure your system 02
☐ Revisit	☐ Done	Configure your system 03
☐ Revisit	☐ Done	Configure your system 04
☐ Revisit	☐ Done	Configure your system 05
☐ Revisit	☐ Done	Configure your system 06
☐ Revisit	☐ Done	Configure your system 07
☐ Revisit	☐ Done	Configure your system 08
☐ Revisit	☐ Done	Configure your system 09
☐ Revisit	☐ Done	Configure your system 10
☐ Revisit	☐ Done	Configure your system 11
☐ Revisit	☐ Done	Configure your system 12
☐ Revisit	☐ Done	Configure your system 13
☐ Revisit	☐ Done	Configure your system 14

Perform the following tasks on node2.domain14.example.com.

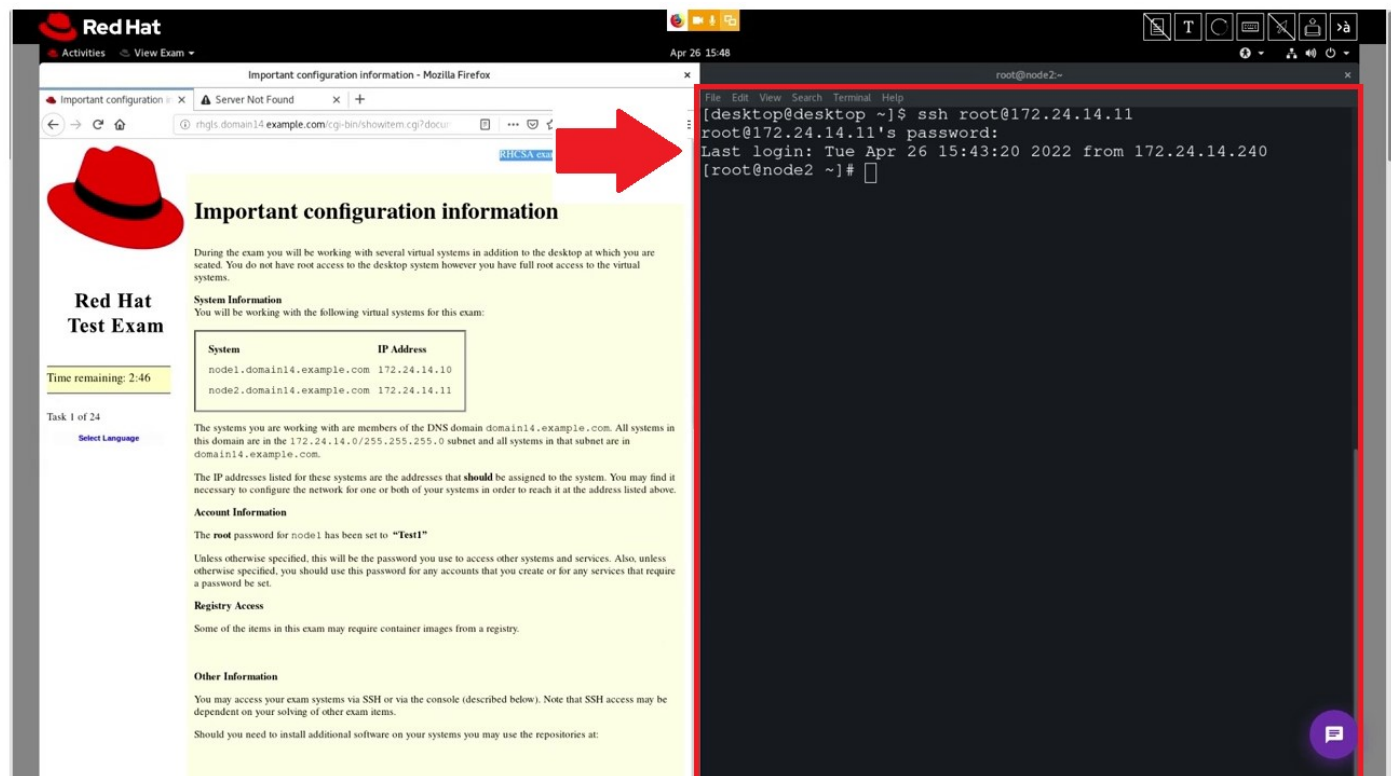
☐ Revisit	☐ Done	Configure your system 15
☐ Revisit	☐ Done	Configure your system 16
☐ Revisit	☐ Done	Configure your system 17
☐ Revisit	☐ Done	Configure your system 18
☐ Revisit	☐ Done	Configure your system 19
☐ Revisit	☐ Done	Configure your system 20
☐ Revisit	☐ Done	Configure your system 21



VM Monitor



Terminal



RHCSA Questions

node 1

No IP address

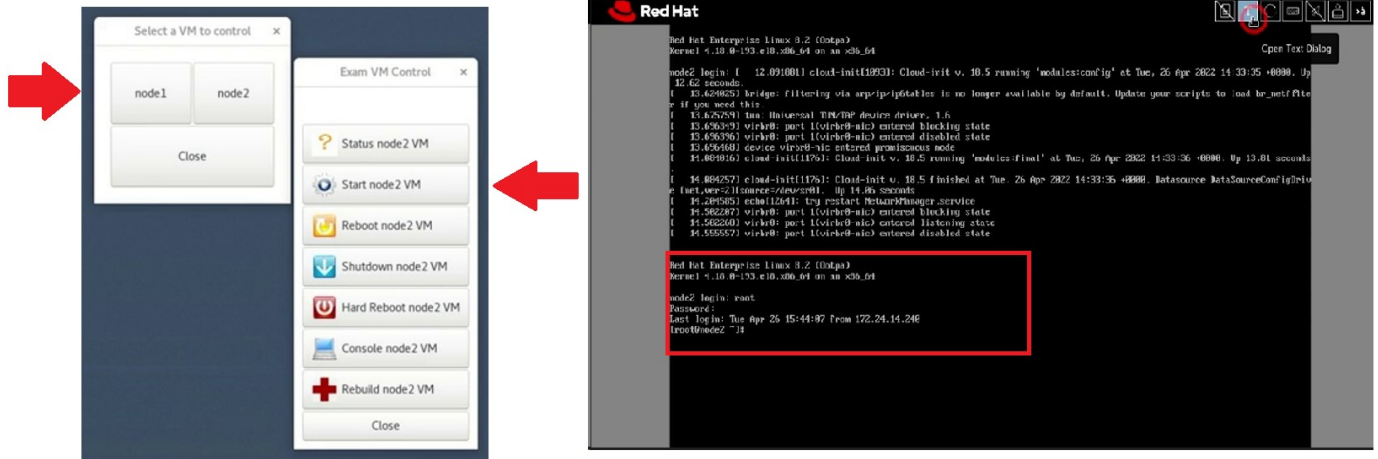
node 2

Root password is not known

Root password already set

IP already set

Perform the question using node-1



Question-1

Configure your Host Name, IP Address, Gateway and DNS.

Host name: station.domain40.example.com

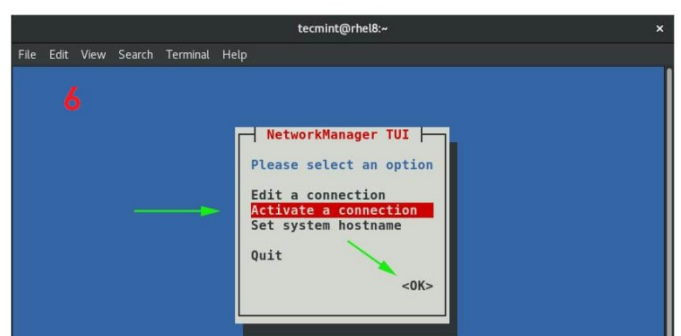
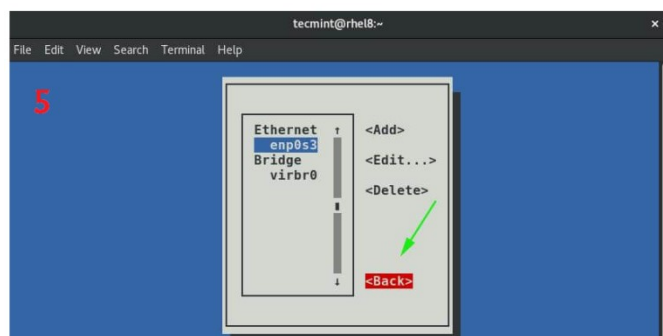
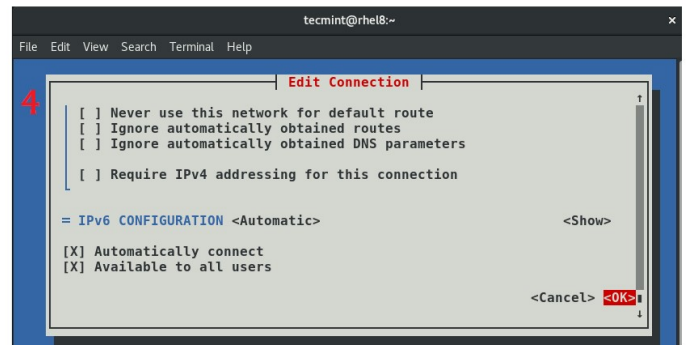
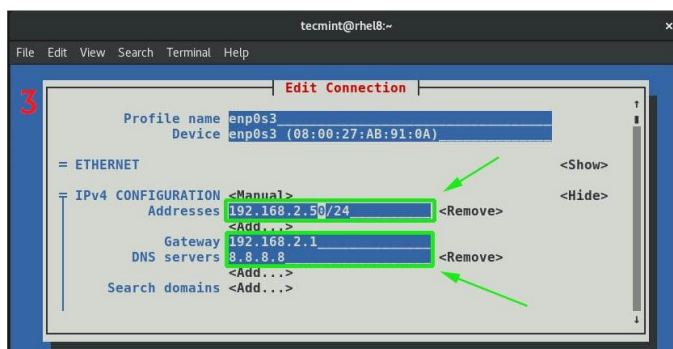
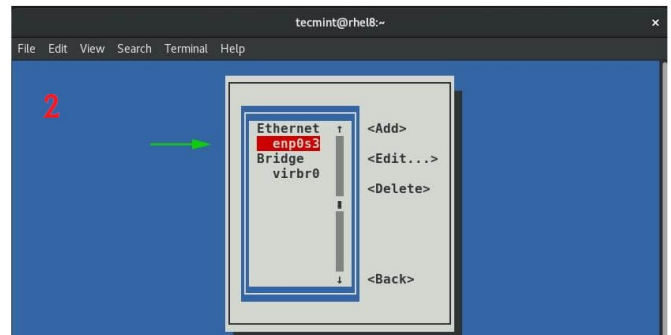
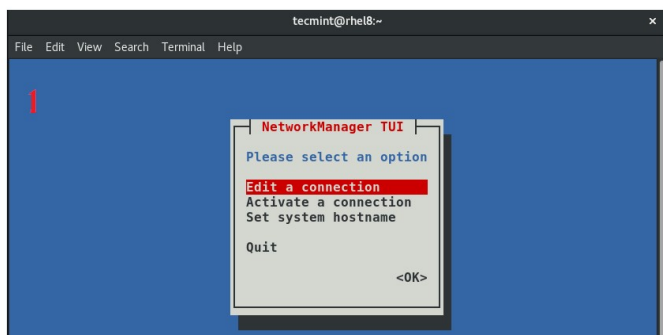
IP Address: 172.24.40.40/24

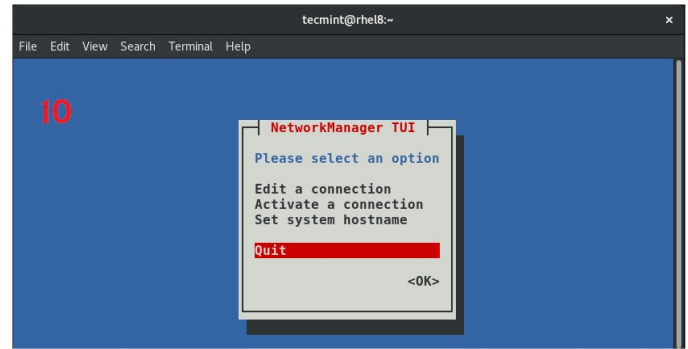
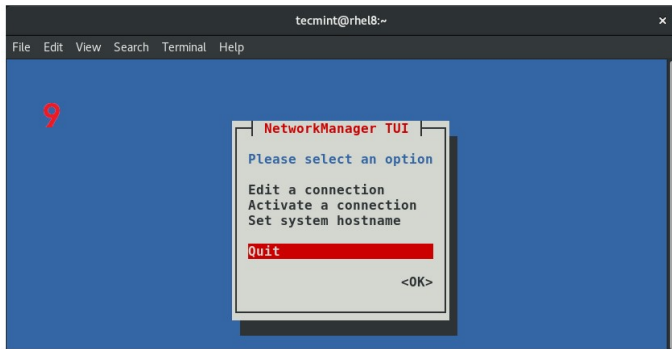
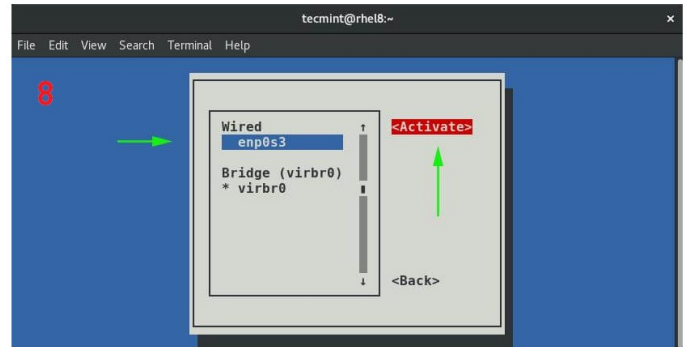
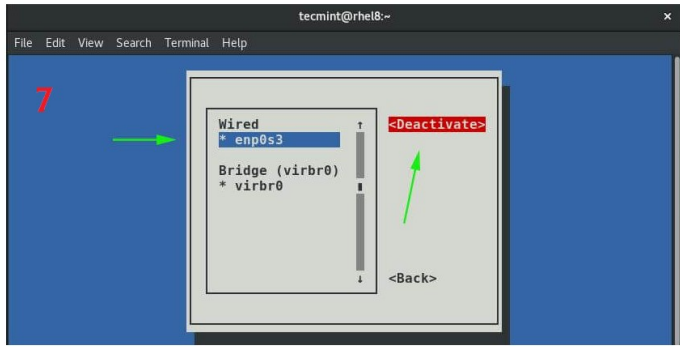
Gateway: 172.24.40.1

DNS/name server: 172.24.40.1

Answer

nmtui





Verification

ip addr show

ping <gateway_ip>

☆ This is a very important part and only after it is completed you will be able to proceed further

Question-2

Configure yum client by using the Yum baseurl paths

http://content.example.com/rhel9.0/x86_64/dvd/BaseOS

http://content.example.com/rhel9.0/x86_64/AppStream

Answer

#yum repolist all (to verify the current status)

#vim /etc/yum.repos.d/yum.repo

[Repo1]

name=BASE REPO

baseurl=http://content.example.com/rhel9.0/x86_64/dvd/BaseOS

enabled=1

gpgcheck=0

[Repo2]

name=APP REPO

baseurl=http://content.example.com/rhel9.0/x86_64/AppStream

enabled=1

gpgcheck=0

:wq!

#yum clean all

#yum repolist all

***Verify the Repo Configuration Result by installing any package in machine Eg: #yum install -y httpd**

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Question-3

The system consists of a web server which serves the contents in /var/www/html. The web server serves output using the port 82, not the default one. Make necessary configurations in your system for serving the content using the port 82

Answer

```
# semanage port -l | grep http
```

```
#semanage port -a -t http_port_t -p tcp 82
```

```
#firewall-cmd-permanent -add-port=82/tcp
```

```
#firewall-cmd -reload
```

```
#systemctl restart httpd
```

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Question-4

Create a group named “admingroup”

- Create a user natasha who have admingroup as supplementary group.
- Create a user harry who have admingroup as a supplementary group.
- Create User sarah should not have access to interactive shell and she should not be a member of “admingroup”
- Set the password of the above three users as “rollis”

Answer

```
#groupadd admingroup
```

```
#useradd -G admingroup natasha
```

```
#useradd -G admingroup harry
```

```
#useradd -s /sbin/nologin sarah
```

```
#passwd natasha
```

```
#passwd harry
```

```
#passwd sarah
```

Question-5

Copy the file /etc/fstab to /var/tmp/fstab and configure the ACL as mentioned below..

- The file is owned by the user root.
- The file belongs to the group root.
- The file should not be executable by anyone.
- The user natasha should be able to read and write to the file.
- The user harry can neither read nor write to the file.
- Other users(future or current) should be able to read the file.

Answer

```
#cp /etc/fstab /var/tmp/fstab
```

```
#chown root:root /var/tmp/fstab
```

```
#chmod ugo-x /var/tmp/fstab
```

```
#setfacl -m u:natasha:rw /var/tmp/fstab
```

```
#setfacl -m u:harry: - /var/tmp/fstab
```

```
#chmod o+r /var/tmp/fstab
```

Question-6

- Create a directory “/home/admins” with the following characteristics.
- Group ownership of “/home/admins” should go to “admingroup”.
- The directory should have read, write and access permission for all the members of “admingroup” group but not to any other users. (It is noted that the “root” has full access to all files present in the system).
- Files created under “/home/admins” should get the same group ownership is set to the “admingroup” group.

Answer

#mkdir /home/admins

#chgrp admingroup /home/admins

#chmod g+rw /home/admins

#chmod o-rwx /home/admins

#chmod g+s /home/admins

Question-7

Configure your machine to be a NTP client of 172.25.254.254

Answer

#timedatectl! (to verify the current status of synchronization in machine)

#timedatectl set-ntp true

#yum install -y chrony

#systemctl start chronyd

#systemctl enable chronyd

#vim /etc/chrony.conf

server 172.25.254.254

#systemctl restart chronyd iburst

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Question-8

The user natasha must configure a cron job that runs at every 2 minutes and execute the job “logger “EX200 EXAM IS ON PROGRESS”.

Answer

```
#yum install -y cronic
```

```
#systemctl start crond
```

```
#systemctl enable crond
```

```
#crontab -eu Natasha
```

```
*/2 * * * * logger “EX200 EXAM IS ON PROGRESS”
```

```
#systemctl restart crond
```

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Question-9

Create a user “alex” with uid 3456 with password “wakennym”.

Answer

```
#useradd -u 3456 alex
```

```
#passwd alex
```



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Question-10

Find the matches with string “squid” from /usr/share/dict/linux.words file and copy the lines in /root/lists.txt file.

Answer

```
#grep 'squid' /usr/share/dict/linux.words >> /root/lists.txt
```



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Question-11

Compression and archive for /usr/local folder as /root/backup.tar.bz2 format.

Answer

#tar cjf /root/backup.tar.bz2 /usr/local



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Question-12

Locate the files of owner “natasha” and copy to the location /root/findresult directory

Answer

```
#mkdir -p /root/findresult
```

```
#find / -type f-user natasha -exec cp -rpvf {} /root/findresult \;
```

Question-13

1. The home directory of LDAP users are shared via NFS.
2. The 172.25.250.10 shares the home directory of ldapusers via NFS.
3. Mount /home/guests/ldapuser1 to your system
4. The ldapuser1 home directory is at **172.25.250.10:/home/guests/ldapuser1**
5. The ldapuser1 home directory should be automounted locally beneath **/home/guests as /home/guests/ldapuser1**
6. The home directories must be writable by their users.
7. For remote logging to the ldapuser using the password "kerberos"

NB: In this question server related configurations are already configured in the machine in the exam console. But Server related configurations and Client related configurations are explained in this doc for your pre exam practice. You need to only configure the client configuration in Real Exam environment

Server configuration

```
#yum install -y nfs-utils
```

```
#systemctl start nfs-server
```

```
#systemctl enable nfs-server
```

```
#mkdir -p /home/guests/ldapuser1
```

#vim /etc/exports

/home/guests/ldapuser1 172.25.250.11(rw)

#exportfs -rv

#systemctl restart nfs-server

#chown nobody /home/guests/ldapuser1

#firewall-cmd --permanent --add-service=nfs

#firewall-cmd --permanent --add-service-rpc-bind

**#firewall-cmd --permanent --add-service-mountd #firewall-cmd --
reload**

Client Configuration

#yum install -y autofs

#systemctl start autofs

#systemctl enable autofs

#vim /etc/auto.master.d/new.autofs

/home/guests /etc/auto.misc

#vim /etc/auto.misc

ldapuser1 -fstype=nfs,rw 172.25.250.10:/home/guests/ldapuser1

#systemctl restart autofs

#df-Th

#cd /home

#ls

#cd guests

#ls

#cd ldapuser1

#touch testfile(For verify the output)



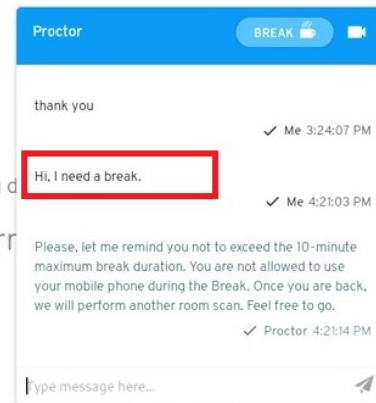
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Break

Your break has begun. Please be aware that the exam clock will continue running during this time.

Please contact your proctor when you have returned.



Break

Your break has begun. Please be aware that the exam clock will continue running during this time.

Please contact your proctor when you have returned.



Perform the question using node-2

Red Hat Certified System Administrator Exam - Mozilla Firefox

Perform the following tasks on node1.domain14.example.com.

- ☐ Revisit ☐ Done [Configure your system 01](#)
- ☐ Revisit ☐ Done [Configure your system 02](#)
- ☐ Revisit ☐ Done [Configure your system 03](#)
- ☐ Revisit ☐ Done [Configure your system 04](#)
- ☐ Revisit ☐ Done [Configure your system 05](#)
- ☐ Revisit ☐ Done [Configure your system 06](#)
- ☐ Revisit ☐ Done [Configure your system 07](#)
- ☐ Revisit ☐ Done [Configure your system 08](#)
- ☐ Revisit ☐ Done [Configure your system 09](#)
- ☐ Revisit ☐ Done [Configure your system 10](#)
- ☐ Revisit ☐ Done [Configure your system 11](#)
- ☐ Revisit ☐ Done [Configure your system 12](#)
- ☐ Revisit ☐ Done [Configure your system 13](#)
- ☐ Revisit ☐ Done [Configure your system 14](#)

Perform the following tasks on node2.domain14.example.com.

- ☐ Revisit ☐ Done [Configure your system 15](#)
- ☐ Revisit ☐ Done [Configure your system 16](#)
- ☐ Revisit ☐ Done [Configure your system 17](#)
- ☐ Revisit ☐ Done [Configure your system 18](#)
- ☐ Revisit ☐ Done [Configure your system 19](#)
- ☐ Revisit ☐ Done [Configure your system 20](#)
- ☐ Revisit ☐ Done [Configure your system 21](#)

A large red arrow points to the 'Configure your system 21' task under node2.

Red Hat VM Control

Select a VM to control

- node1
- node2**

Exam VM Control

- Status node2 VM
- Start node2 VM
- Reboot node2 VM**
- Shutdown node2 VM
- Hard Reboot node2 VM
- Console node2 VM
- Rebuild node2 VM

GLS VM Console - Mozilla Firefox

```
Red Hat Enterprise Linux 8.2 (otopa)
Kernel 4.18.0-193.el8.x86_64 on an x86_64

node2 login: [ 12.891881] cloud-init[1193]: Cloud-init v. 18.5 running 'modules:final' at Tue, 26 Apr 2022 14:33:35 +0000. Up
12.62 seconds.
[ 13.624825] bridge: filtering via arp/ip6tables is no longer available by default. Update your scripts to load br_netfilter
if you need this.
[ 13.675753] tun: Universal TUN/TAP device driver, 1.6
[ 13.696349] virbr0: port 1(virbr0-nic) entered blocking state
[ 13.696396] virbr0: port 1(virbr0-nic) entered disabled state
[ 13.696468] device virbr0-nic entered promiscuous mode
[ 14.004016] cloud-init[1176]: Cloud-init v. 18.5 running 'modules:final' at Tue, 26 Apr 2022 14:33:36 +0000. Up 13.81 seconds
[ 14.004257] cloud-init[1176]: Cloud-init v. 18.5 finished at Tue, 26 Apr 2022 14:33:36 +0000. DataSource DataSourceConfigDrive
net.net-2[1193]: Up 14.86 seconds
[ 14.298365] echo[1264]: try restart NetworkManager.service
[ 14.582287] virbr0: port 1(virbr0-nic) entered blocking state
[ 14.582288] virbr0: port 1(virbr0-nic) entered listening state
[ 14.555557] virbr0: port 1(virbr0-nic) entered disabled state
```

Root password change

```
Red Hat Enterprise Linux (5.14.0-70.13.1.el9_0.x86_64) 9.0 (Plow)
Red Hat Enterprise Linux (0-rescue-c4d3dca7b2c14b5abf5e1159d64881a0) 9.0-

1

Use the ↑ and ↓ keys to change the selection.
Press 'e' to edit the selected item, or 'c' for a command prompt.
```

```
load_video
set gfxpayload=keep
insmod gzio
linux ($root)/vmlinuz-5.14.0-70.13.1.el9_0.x86_64 root=/dev/mapper/rhel-root r\
o crashkernel=16-46:192M,46-646:256M,646-:512M resume=/dev/mapper/rhel-swap rd\
.lvm.lv=rhel/root rd.lvm.lv=rhel/swap rhgb quiet rd.break ← 2
initrd ($root)/initramfs-5.14.0-70.13.1.el9_0.x86_64.img

Press Ctrl-x to start, Ctrl-c for a command prompt or Escape to
discard edits and return to the menu. Pressing Tab lists
possible completions.
```

```
Generating "/run/initramfs/rdsosreport.txt"
3
Entering emergency mode. Exit the shell to continue.
Type "journalctl" to view system logs.
You might want to save "/run/initramfs/rdsosreport.txt" to a USB stick or /boot
after mounting then and attach it to a bug report.

Press Enter for maintenance
(or press Control-D to continue):
sh-5.1#
```

mount -o remount,rw /sysroot

4 # chroot /sysroot

passwd root

```
sh-5.1# passwd
Changing password for user root.
New password:
Retype new password: 5
passwd: all authentication tokens updated successfully.
sh-5.1# _
```

touch /.autorelabel

6 # exit

exit

Question-1

Configure yum client by using the Yum baseurl paths

http://content.example.com/rhel9.0/x86_64/dvd/BaseOS

http://content.example.com/rhel9.0/x86_64/AppStream

Answer

#yum repolist all (to verify the current status)

#vim /etc/yum.repos.d/yum.repo

[Repo1]

name=BASE REPO

baseurl=http://content.example.com/rhel9.0/x86_64/dvd/BaseOS

enabled=1

gpgcheck=0

[Repo2]

name=APP REPO

baseurl=http://content.example.com/rhel9.0/x86_64/AppStream

enabled=1

gpgcheck=0

:wq!

#yum clean all

#yum repolist all



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Question-2

Resize the lvm size to 300MiB the location from "/dev/vg/vo" without losing any data. Size between 270 MiB and 330MiB is permissible

Answer

#lvs (To check the current size of /dev/vg/vo)

#df-Th (To check the file system used for formatting)

If the file system is 'ext4', do the following

#lvextend -L 300M /dev/vg/vo

#resize2fs /dev/vg/vo

If the file system is 'xfs', do the following

#lvextend -L 300M /dev/vg/vo

#xfs_growfs <mount point>

Question-3

Add the swap space with “512”MiB on your system. Don’t remove the existing swap. Your new swap should be mounted at booting time also.

Answer

#free -mt (To check the current status of swap memory in machine)

#fdisk /dev/vdb

n

p

Select partition number: 2

First sector: <Press Enter>

Last sector: +512M

t

82

W

#partprobe

#mkswap /dev/vdb2


```
#vim /etc/fstab
```

```
/dev/vdb2 swap swap defaults 0 0
```

```
:wq!
```

```
#swapon -a
```

```
#free -mt (verify the output)
```



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Question-4

Create the “LVM” with the name “database” by using 50PE’s from the volume group “datastore”. Consider the PE size as “16MB”. Mount it on /mnt/database with file system **ext3**.

Answer

#lsblk (To check the free space in disk drives given in machine)

#fdisk /dev/vda

n

p

Select partition number: 3

First sector: <Press Enter> Last sector: +1G

t

8e

w

#partprobe

#pvcreate /dev/vda3

#vgcreate -s 16M datastore /dev/vda3

#lvcreate -L 800M -n database datastore

#mkfs.ext3 /dev/datastore/database

```
#mkdir /mnt/database
```

```
#vim /etc/fstab
```

```
/dev/datastore/database /mnt/database ext3 defaults 0 0
```

```
:wq!
```

```
#mount -a
```

```
#df-Th (To check the result)
```



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Question-5

Activate the tuning profile in your system which is recommended by the system by default.

Answer

```
#yum install -y tuned
```

```
#systemctl start tuned
```

```
#systemctl enable tuned
```

```
#tuned-adm recommend
```

```
#tuned-adm profile <profile name getting from the previous  
command>
```

```
#tuned-adm active
```

Question-6

- Create a new customized environment for your users.
- Create a new custom command called "userstat.sh" whose output should be: command "hostname"
- Make sure the "userstat" command should be available by-default for all users

Answer

```
#vim userstat.sh
```

```
#!/bin/bash
```

```
hostname
```

```
:wq!
```

```
#chmod +x userstat.sh
```

```
#cp userstat.sh /usr/bin
```

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Question-7

Create a shell script locate.sh whose output should be a directory named /root/locateresult which contains all the files of owner "bob"

Answer

```
#vim locate.sh
```

```
#!/bin/bash
```

```
mkdir /root/locateresult
```

```
find/-type f-user bob -exec cp -rpvf {} /root/locate result \;
```

```
:wq!
```

```
#chmod +x locate.sh
```

```
#cp locate.sh /usr/bin
```

```
#locate.sh
```

Container Question-1

Build a container image named 'monitor' from the file <http://classroom.lab.example.com/containerfile> as user 'seshta'

Container Question-2

- Create a rootless detached container named 'ascii2pdf' from the previously created 'monitor' image.
- Create a service named 'container-ascii2pdf.service' from the above container.
- Make the container with the /opt/files folder on the host mounted at /opt/incoming on the container.
- Make the container with the /opt/processed folder on the host mounted at /opt/outgoing on the container.
- The container should start automatically when the system starts.

NOTE: If the service is correctly configured, for any plain text placed in /opt/files folder, a pdf with the same name will be created inside /opt/outgoing in container

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Container Solution – Both questions

As Root:

Container-tools were already installed. If not, install with

Yum install container-tools -y

Mkdir /opt/files

Mkdir /opt/processed

Chown seshta:seshta /opt/files

Chown seshta:seshta /opt/processed

Logout of node1 and log back in as user 'seshta'

ssh seshta@<hostname>

Wget <http://classroom.lab.example.com/containerfile>

Podman build -t monitor:1.0 .

Podman images

Podman create -d --name ascii2pdf -v /opt/files:/opt/incoming:Z -v /opt/processed:/opt/outgoing:Z localhost/monitor:1.0

Podman start ascii2pdf

Podman ps

Mkdir .config/systemd/user


```
# Cd .config/systemd/user
# Podman generate systemd --name ascii2pdf --files --new
# ls
# Podman stop ascii2pdf
# Podman rm ascii2pdf
# Podman ps -a
# Systemctl --user daemon-reload
# Systemctl --user enable --now container-ascii2pdf.service
# Podman ps
# Loginctl enable-linger
# Loginctl show-user seshta
```

To check the working of container:

```
# echo "Hello World" >> /opt/files/test
# Podman exec -it ascii2pdf /bin/bash
# ls /opt/incoming
# ls /opt/outgoing
```

press ctrl+D to exit from container